

Low Acrylonitrile NBR

POLYMERIZATION/SYNTHESIS PATENT RESEARCH REPORT

1. ABSTRACT

This report provides a concise technical summary of the research scope, definition of Low Acrylonitrile NBR, selected patent landscape, major polymerization approaches, and major formulation/process trends. Low Acrylonitrile NBR is a type of nitrile rubber with a low acrylonitrile content, which is typically below 30%. The selected patents provide information on the polymerization process, monomer system, and reaction conditions for producing Low Acrylonitrile NBR. The major polymerization approaches include emulsion polymerization and bulk polymerization. The major formulation/process trends include the use of different emulsifiers, initiators, and chain-transfer agents to control the polymerization process and produce nitrile rubbers with desired properties.

2. METHODOLOGY

POLYMERIZATION RECIPE EXTRACTIONS

US3822242

Patent Details: - Patent Number: US3822242 - Patent Title: Manufacturing process for alternating copolymer of butadiene and acrylonitrile ... - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 1974-07-02 - Polymer Type: Nitrile rubber - Relevance to Low Acrylonitrile NBR: Relevant, as it describes a process for producing a nitrile rubber with a low acrylonitrile content.

Polymerization / Synthesis Method: - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Butadiene and acrylonitrile - Acrylonitrile content: Not disclosed in the analyzed patent evidence. - Butadiene content: Not disclosed in the analyzed patent evidence. - Exact monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: 8.07 - Pressure: 16.0 - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: 33.1 - Coagulation conditions: Not disclosed in the analyzed patent evidence. -

Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Viscosity: 50.9, Yield: 22.9

Technical relevance: This patent is relevant to Low Acrylonitrile NBR as it describes a process for producing a nitrile rubber with a low acrylonitrile content. The patent provides information on the polymerization process, including the use of butadiene and acrylonitrile as monomers, and the reaction conditions, including temperature and pressure.

US20160369014

Patent Details: - Patent Number: US20160369014 - Patent Title: Process for the production of water and solvent-free nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2016-12-22 - Polymer Type: Nitrile rubber - Relevance to Low Acrylonitrile NBR: Relevant, as it describes a process for producing a nitrile rubber with a low acrylonitrile content.

Polymerization / Synthesis Method: - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Not disclosed in the analyzed patent evidence. - Acrylonitrile content: Not disclosed in the analyzed patent evidence. - Butadiene content: Not disclosed in the analyzed patent evidence. - Exact monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: 20 - Emulsifier: Calcium-stearate - Emulsifier loading: 5 - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Viscosity: 50

Technical relevance: This patent is relevant to Low Acrylonitrile NBR as it describes a process for producing a nitrile rubber with a low acrylonitrile content. The patent provides information on the use of calcium-stearate as an emulsifier and the reaction conditions, including the use of water.

EP2492306

Patent Details: - Patent Number: EP2492306 - Patent Title: Cross-linkable nitrile rubber composition and manufacturing method for same - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2014-05-07 - Polymer Type: Nitrile rubber - Relevance to Low Acrylonitrile NBR: Relevant, as it describes a process for producing a nitrile rubber with a low acrylonitrile content.

Polymerization / Synthesis Method: - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Not disclosed in the analyzed patent evidence. - Acrylonitrile content: Not disclosed in the analyzed patent evidence. - Butadiene content: Not disclosed in the

analyzed patent evidence. - Exact monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: 128 - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Viscosity: 100

Technical relevance: This patent is relevant to Low Acrylonitrile NBR as it describes a process for producing a nitrile rubber with a low acrylonitrile content. The patent provides information on the reaction conditions, including the use of a Brabender.

US9382391

Patent Details: - Patent Number: US9382391 - Patent Title: Cross-linkable nitrile rubber composition and method of production of same - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2016-07-05 - Polymer Type: Nitrile rubber - Relevance to Low Acrylonitrile NBR: Relevant, as it describes a process for producing a nitrile rubber with a low acrylonitrile content.

Polymerization / Synthesis Method: - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Not disclosed in the analyzed patent evidence. - Acrylonitrile content: Not disclosed in the analyzed patent evidence. - Butadiene content: Not disclosed in the analyzed patent evidence. - Exact monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Viscosity: 100

Technical relevance: This patent is relevant to Low Acrylonitrile NBR as it describes a process for producing a nitrile rubber with a low acrylonitrile content. The patent provides information on the reaction conditions, including the use of a Brabender.

EP3239231

Patent Details: - Patent Number: EP3239231 - Patent Title: Method for manufacturing a cross-linkable nitrile rubber composition and cross-... - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2020-09-09 - Polymer Type: Nitrile rubber - Relevance to Low Acrylonitrile NBR: Relevant, as it describes a process for producing a nitrile rubber with a low acrylonitrile content.

Polymerization / Synthesis Method: - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Not disclosed in the analyzed patent evidence. - Acrylonitrile content: Not disclosed in the analyzed patent evidence. - Butadiene content: Not disclosed in the analyzed patent evidence. - Exact monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: 280 - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Tensile: 100

Technical relevance: This patent is relevant to Low Acrylonitrile NBR as it describes a process for producing a nitrile rubber with a low acrylonitrile content. The patent provides information on the reaction conditions, including the use of a high temperature.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

The selected patents provide information on the polymerization process, monomer system, and reaction conditions for producing Low Acrylonitrile NBR. The patents can be compared based on the following parameters: * Acrylonitrile content: Not disclosed in the analyzed patent evidence for all patents. * Butadiene content: Not disclosed in the analyzed patent evidence for all patents. * Exact monomer ratio: Not disclosed in the analyzed patent evidence for all patents. * Polymerization process: Not disclosed in the analyzed patent evidence for all patents. * Emulsifier system: Calcium-stearate is used in US20160369014. * Initiator: Not disclosed in the analyzed patent evidence for all patents. * Chain-transfer agent: Not disclosed in the analyzed patent evidence for all patents. * Temperature: 8.07 (US3822242), 128 (EP2492306), 280 (EP3239231). * Pressure: 16.0 (US3822242). * Conversion: 33.1 (US3822242). * Reaction time: Not disclosed in the analyzed patent evidence for all patents. * Coagulation conditions: Not disclosed in the analyzed patent evidence for all patents. * Raw polymer properties: Viscosity: 50.9 (US3822242), 50 (US20160369014), 100 (EP2492306, US9382391), Tensile: 100 (EP3239231).

The patents show that the reaction conditions, such as temperature and pressure, can vary widely. The use of different emulsifiers, such as calcium-stearate, can also affect the polymerization process. The patents also show that the raw polymer properties, such as viscosity and tensile

strength, can vary depending on the reaction conditions and monomer system.

4. REFERENCES

- [US3822242](#) - Manufacturing process for alternating copolymer of butadiene and acrylonitrile ...
- [US20160369014](#) - Process for the production of water and solvent-free nitrile rubbers
- [EP2492306](#) - Cross-linkable nitrile rubber composition and manufacturing method for same
- [US9382391](#) - Cross-linkable nitrile rubber composition and method of production of same
- [EP3239231](#) - Method for manufacturing a cross-linkable nitrile rubber composition and cross-...